Program 7: Deploy a Web Application to Kubernetes using Minikube

Project Structure

program7
|--nginx-deployment.yaml

Step 1: Start the minikube

```
rv24mc053_kavya@kavya-Victus-by-HP-Gaming-Laptop-15-fa1xxx:~$ minikube start
😄  minikube v1.37.0 on Ubuntu 24.04
✨  Using the docker driver based on existing profile
👍  Starting "minikube" primary control-plane node in "minikube" cluster
🚜  Pulling base image v0.0.48 ...
🔄  Restarting existing docker container for "minikube" ...
❗  Failing to connect to https://registry.k8s.io/ from inside the minikube container
💡  To pull new external images, you may need to configure a proxy: https://minikube.sigs.k8s.io/docs/reference/networking/proxy/
🐳  Preparing Kubernetes v1.34.0 on Docker 28.4.0 ...
🔎  Verifying Kubernetes components...
    ▪ Using image docker.io/kubernetesui/dashboard:v2.7.0
    ▪ Using image docker.io/kubernetesui/metrics-scraper:v1.0.8
    ▪ Using image registry.k8s.io/metrics-server/metrics-server:v0.8.0
    ▪ Using image gcr.io/k8s-minikube/storage-provisioner:v5
💡  Some dashboard features require the metrics-server addon. To enable all features please run:

        minikube addons enable metrics-server

🌟  Enabled addons: storage-provisioner, default-storageclass, metrics-server, dashboard
🏄  Done! kubectl is now configured to use "minikube" cluster and "default" namespace by default
```

Step 2: Check the status of minikube

```
rv24mc053_kavya@kavya-Victus-by-HP-Gaming-Laptop-15-fa1xxx:~$ minikube status
minikube
type: Control Plane
host: Running
kubelet: Running
apiserver: Running
kubeconfig: Configured
```

Step 3: Create deployment file

nano nginx-deployment.yaml

Step 4: Apply the nginx-deployment.yaml

```
rv24mc053_kavya@kavya-Victus-by-HP-Gaming-Laptop-15-fa1xxx:~$ nano nginx-deployment.yml
rv24mc053_kavya@kavya-Victus-by-HP-Gaming-Laptop-15-fa1xxx:~$ kubectl apply -f nginx-deployment.yml
deployment.apps/nginx-deployment created
```

Step 5: Check status

```
rv24mc053_kavya@kavya-Victus-by-HP-Gaming-Laptop-15-fa1xxx:~$ kubectl get deployments
NAME               READY   UP-TO-DATE   AVAILABLE   AGE
nginx-deploy       3/3     3            3           10d
nginx-deployment   0/2     2            0           20s
```

Step 6: Expose the deployment as a Service

```
rv24mc053_kavya@kavya-Victus-by-HP-Gaming-Laptop-15-fa1xxx:~$ kubectl expose deployment nginx-deployment \
  --type=NodePort \
  --port=80
service/nginx-deployment exposed
```

Step 7: Check service

```
rv24mc053_kavya@kavya-Victus-by-HP-Gaming-Laptop-15-fa1xxx:~$ kubectl get svc
NAME               TYPE           CLUSTER-IP       EXTERNAL-IP   PORT(S)          AGE
kubernetes         ClusterIP      10.96.0.1        <none>        443/TCP          16d
kubia              NodePort       10.102.41.198    <none>        8080:31106/TCP   3d23h
kubia-http         LoadBalancer   10.101.20.228    <pending>     8080:31284/TCP   3d23h
nginx-deploy       ClusterIP      10.96.67.146     <none>        80/TCP           10d
nginx-deployment   NodePort       10.105.162.53    <none>        80:32104/TCP     16s
nginx-pod          ClusterIP      10.99.24.33      <none>        80/TCP           10d
```

Step 8: Access the Web Application

```
rv24mc053_kavya@kavya-Victus-by-HP-Gaming-Laptop-15-fa1xxx:~$ minikube service nginx-deployment
|-----------|------------------|-------------|---------------------------|
| NAMESPACE |       NAME       | TARGET PORT |            URL            |
|-----------|------------------|-------------|---------------------------|
| default   | nginx-deployment | 80          | http://192.168.49.2:32104 |
|-----------|------------------|-------------|---------------------------|
🎉  Opening service default/nginx-deployment in default browser...
rv24mc053_kavya@kavya-Victus-by-HP-Gaming-Laptop-15-fa1xxx:~$ Gtk-Message: 15:42:28.164: Not loading module "atk-bridge": The functionality is provided by GTK native
ot load it.
```